

Special section VLCC – Lagrangian V10, Tenv, Continuous morphogenic attractors and open variational structure

From relativistic time to variational time?

Frederick Vronsky

Independent researcher in ORCID theoretical cosmology:

<https://orcid.org/0009-0003-5719-9604> License: Creative

Commons BY-NC-SA 4.0 International Toulouse – May 2026

Invitation to a broader reading

The present work introduces the V10 variational framework of the VLCC model and marks an important evolution of its operational architecture, notably through the introduction of a continuous attractorial reading of morphogenic regimes and through the stabilization of several variational structures of the model.

Readers wishing to delve deeper into the historical, conceptual, and mathematical foundations of the VLCC may refer to:

VLCC Treatise — Canonical Edition, Version 1

DOI (Zenodo):

<https://doi.org/10.5281/zenodo.17946156>

This document brings together the main formulations that led to the V10 framework presented in this work.

Table of Contents

Chapter I — VLCC V10 OPERATIONAL CORE	3
1.1 —VLCC V10 operative introduction	3
1.2 —Three-phase field and chronotropic structure	3
1.3 —Morphogenic invariant and chronotropic polarity	4
1.4 —Morphogenic weighting and operational calibration	5
Chapter II — MORPHOGENIC SPACE AND CONTINUOUS ATTRACTORS	7
2.1 —Morphogenic space and attractor coordinates	7
2.2 —Operational projection of morphogenic space	7
2.3 —Morphogenic gradation and continuous attractors	9
2.4 —Morphogenic Variator and Attractiveness Weighting	9
2.5 —Morphogenic slippage and coherence transitions	11
Chapter III — MORPHOGENIC INERTIAL ENVIRONMENT	12
3.1 —Definition and operational status of the inertial environment (Tenv)	12
3.2 —Effective inertial responses and morphogenic mass	12
3.3 —Effective energy response and inertial reformulation	13
Chapter IV —VARIATIONAL AND LAGRANGIAN STRUCTURE V10	14
4.1 —variational condition and open framework equation	14
4.2 —Attractive variational contributions	15
4.3 —Variational Lagrangian V10	16
4.4 —Explicit morphogenic memory	18
4.5 —VLCC V10 Operating Architecture	19
Chapter V —REVEABILITY AND MORPHOGENEIC LIMITS	22
5.1 —Morphogenic revealability and observable coherence	22
5.2 —Residual coherence and asymptotic structures	23
Chapter VI —OPERATIONAL EXTENSIONS AND MODEL STATUS	26
6.1 —Morphogenic continuity and cosmological structures	26
6.2 —Operative status and limitations of the VLCC V10	27
Annex A —Lexicon	32
Annex B —Detailed equations	38
Annex C —Table of morphogenic attractors	40
Appendix D —Mathematical Glossary	42

CHAPTER I — VLCC V10 OPERATIONAL CORE

I.1 — Operative positioning of the VLCC V10

The VLCC (Variable Lagrangian of Cosmic Chronotropy) model proposes a speculative variational framework intended to describe certain dynamics of temporal coherence and their morphogenic, inertial and cosmological manifestations.

Previous versions of the model introduced:

- the three-phase field,
- the morphogenic triplet,
- the sliding mechanisms,
- morphogenic gradients,
- as well as the first variational structures of the VLCC.

The present work primarily aims at the operational stabilization of the V10 frame.

This reformulation introduces several important changes:

- a continuous reading of the old morphogenic gradients;
- an attractor space defined in the plane (χ_1, χ_3) ;
- a morphogenic variator ensuring the continuous weighting of attractors;
- an operational reformulation of the inertial environment Tenv;
- an open variational framework equation;
- and a Lagrangian V10 intended to stabilize local redistributions of morphogenic coherence.

In the VLCC V10, the old gradients are no longer interpreted as rigid discrete regimes but as continuous morphogenic attractors organized in a coherent variational space.

The proposed model is neither a complete theory of quantum gravity nor a closed reformulation of general relativity. The VLCC V10 should be understood as an effective operational framework intended to formalize certain coherence dynamics of the present observable.

The historical, ontological developments and earlier formulations of the model are presented in the Treatise VLCC — Canonical Edition.

I.2 — Three-phase field and chronotropic structure

The VLCC V10 is based on a fundamental three-phase field:

$$T = (\tau, \sigma, \Delta t)$$

Or :

- τ represents morphogenic tension;
- σ local morphogenic coherence;
- Δt the internal chronotropic polarity.

These components describe operational variables of coherence of the present observable and not additional dimensions of spacetime.

The three-phase field is based on an internal dynamic structure defined by the morphogenic triplet:

(t_1, t_2, t_1')

Or :

- t_1 corresponds to stabilized morphogenic memory;
- t_2 in the present coherent operational form;
- t_1' to the evolving tension of the system.

The VLCC interprets the present observable as a dynamic structure maintained by the continuous redistribution of the internal weights of the triplet:

$t_1' \rightarrow t_2 \rightarrow t_1$

Chronotropic polarity is defined by:

$\Delta t = t_1' - t_1$

This quantity acts as a local indicator of morphogenic asymmetry and intervenes directly in sliding mechanisms and attractorial transitions.

Within the operational framework of the VLCC V10, observable structures are interpreted as emergent states of local morphogenic coherence dependent on internal redistributions of the tri-phase field.

Detailed developments relating to the three-phase field and the morphogenic triplet are presented in the VLCC Treatise — Canonical Edition.

I.3 — Morphogenic invariant and chronotropic polarity

The VLCC V10 introduces a morphogenic invariant intended to characterize the local stability conditions of the present observable:

$I = t_2^2 - t_1 t_1'$

The condition:

$$I > 0$$

corresponds to the permissible morphogenic stability domain of the system.

In this approach, chronotropic stability results from a dynamic balance between morphogenic memory, present coherence, and evolutionary tension.

The VLCC thus interprets the chronotropic arrow as an emergent property linked to the continuous redistribution of the coherence structures of the morphogenic triplet rather than as a geometric orientation imposed a priori.

Polarity:

$$\Delta t = t_1' - t_1$$

directly intervenes:

- in sliding mechanisms;
- in attractor transitions;
- and in local redistributions of morphogenic coherence.

The morphogenic invariant thus constitutes one of the fundamental operational conditions of the V10 variational framework.

I.4 — Morphogenic weighting and operational calibration

The VLCC treaty introduced a morphogenic weighting intended to describe the dynamic distribution of the components of the morphogenic triplet within the different coherence regimes.

The weightings associated with:

$$(t_1, t_2, t_1')$$

correspond to effective coherence coefficients describing the local state of the three-phase field.

In this approach, morphogenic regimes result from continuous redistributions of the system's internal weightings rather than from rigid separations between distinct domains.

The formulations of the VLCC treaty introduced calibrations associated with the Planck scale. Within the framework of V10, these relationships are retained as operational tools for normalizing variational structures.

Morphogenic weighting plays a role, in particular:

- in the stabilization of continuous attractors;
- in sliding mechanisms;
- in coherence transitions;
- and in the operational definition of the inertial environment Tenv.

This reformulation allows us to preserve a continuous variational reading of the model compatible with the attractorial dynamics of the VLCC V10.

PART II — MORPHOGENIC SPACE AND CONTINUOUS ATTRACTORS

II.1 — Morphogenic space and attractor coordinates

The VLCC V10 introduces a continuous representation of morphogenic regimes based on an operational space designed to describe the local coherence states of the three-phase field. The old morphogenic gradients are no longer interpreted as discrete domains but as continuous attractors organized within a dynamic morphogenic space.

This reformulation is based on the introduction of a morphogenic plane defined by the operational coordinates:

(χ_1, χ_3)

Or :

- χ_1 mainly describes the level of local morphogenic coherence associated with stabilized structures;
- χ_3 describes the relative level of evolutionary tension and morphogenic slippage.

The coordinates (χ_1, χ_3) constitute operational variables describing the local dynamic position of a system in the morphogenic coherence space of the VLCC.

The morphogenic plane allows us to represent:

- morphogenic attractors;
- transitions between regimes;
- landslide zones;
- hybrid structures;
- local redistributions of coherence;
- and the morphogenic trajectories of the system.

II.2 — Operational projection of morphogenic space

Within the framework of VLCC V10, the morphogenic space:

(χ^1, χ^3)

constitutes a general operational structure intended to represent the local states of morphogenic coherence of the three-phase field. In order to make this structure usable in certain observable astrophysical regimes, the VLCC introduces operational projections linking the morphogenic coordinates to certain local physical quantities.

A minimal formulation is introduced in the form:

$$\chi^1 = \Omega / \sqrt{g_{\text{bar}} \Sigma L}$$

And :

$$\chi^3 = (R \cdot f_{\text{gaz}}) / (\Sigma L \cdot \Omega_{\text{bar}})$$

Or :

- Ω represents a quantity of local rotational coherence;
- g_{bar} represents the effective baryonic contribution;
- ΣL represents the local light surface density;
- R represents the characteristic radial scale of the system;
- f_{gaz} represents the effective gaseous fraction;
- Ω_{bar} represents the local baryonic rotational contribution.

These relations do not constitute direct fundamental observables but operational projections intended to represent certain local distributions of morphogenic coherence in the attractorial space of the VLCC.

This reformulation allows us to connect the morphogenic landscape:

- to local rotational structures;
- to baryon distributions;
- to light density regimes;
- and to the dynamic states of observable systems.

Morphogenic coordinates thus become operational variables capable of representing:

- the dominant attractor regimes;
- morphogenic transitions;
- hybrid structures;
- the redistributions of coherence;
- and the local trajectories of the system in variational space.

Within the framework of VLCC V10, observable structures are no longer associated with separate rigid regimes but with continuous distributions of morphogenic coherence dependent on local attractor weights.

This approach allows, in particular:

- to introduce a continuous reading of cosmological transitions;
- to prepare certain observational confrontations;
- and to stabilize the operational character of the morphogenic plan (χ^1, χ^3) within the variational framework of the VLCC V10.

II.3 — Morphogenic Graduation and Continuous Attractors

Morphogenic attractors are distributed according to three main phases of coherence:

Phase I — Highly coherent regimes G1,
G1.2, G1.3

Phase II — Transitional regimes G2,
G2.1, G2.2

Phase III — Weak coherent regimes G3,
G3.1, G3.2

Each attractor is associated with a particular distribution of internal weights of the morphogenic triplet as well as a specific level of operational coherence.

II.4 — Morphogenic Variator and Attractiveness Weighting

Within the VLCC V10 framework, the morphogenic variator constitutes one of the main dynamic mechanisms of the model. It allows us to describe the continuous weighting of morphogenic attractors within the variational space (χ_1, χ_3) and to interpret observable structures as dynamic superpositions of local attractor contributions.

The attractor weights are described by the functions:

$w_i(r)$

associated with the different G_i attractors of the morphogenic landscape.

In simplified VLCC formulations, the variator could be represented as:

$$w_i(r) = \mu w_i(r - \Delta r) + (1 - \mu) \exp[-d_i(r) / \lambda M]$$

This writing made it possible to introduce an elementary dynamic of morphogenic memory and attractorial redistribution.

However, this formulation remained insufficient to properly describe the continuous normalization of the attractor weights and the overall variational stability of the system.

The VLCC V10 now introduces a standardized canonical formulation of the morphogenic variator:

$$w_i(r) = \mu w_i(r - \Delta r) + (1 - \mu) [\exp(-d_i^2(r) / \Theta M) / \sum_j \exp(-d_j^2(r) / \Theta M)]$$

Or :

- $w_i(r)$ represents the local weighting of the attractor G_i ;
- μ represents the morphogenic memory coefficient;
- $d_i^2(r)$ represents the local morphogenic distance between the system and the attractor G_i ;
- ΘM represents the effective scale of morphogenic diffusion.

This formulation introduces a continuous variational structure based on a normalized redistribution of attractor weights. The variator thus becomes capable of representing:

- progressive morphogenic transitions;
- hybrid regimes;
- local attraction competitions;
- and the mechanisms for the continuous stabilization of the system.

The local morphogenic distance is defined by:

$$d_i^2(r) = a_i (\chi_1 - \chi_{1,i})^2 + b_i (\chi_3 - \chi_{3,i})^2$$

Or :

- a_i and b_i represent the local geometric weighting coefficients;
- $\chi_{1,i}$ and $\chi_{3,i}$ represent the attractor coordinates associated with the attractor G_i .

This distance describes the local relative position of the system in morphogenic space with respect to the continuous attractors of the VLCC.

The term:

$$\mu w_i(r - \Delta r)$$

introduces a morphogenic persistence associated with previously stabilized regimes.

The standardized term:

$$\exp(-d_i^2(r) / \Theta M) / \sum_j \exp(-d_j^2(r) / \Theta M)$$

introduces a continuous redistribution of attractor weights depending on the local morphogenic position of the system.

This formulation helps to preserve:

- variational continuity;
- attractiveness;
- the stability of hybrid regimes;
- and the mechanisms of morphogenic slippage.

The morphogenic variator intervenes directly:

- in the definition of the inertial environment T_{env} ;
- in effective inertial responses;
- in attractor transitions;
- in the mechanisms of revealability;
- and in the variational structure of the Lagrangian V_{10} .

The VLCC V_{10} thus introduces a continuous dynamic of morphogenic redistribution allowing to stabilize the continuous attractors within the variational landscape of the model.

II.5 — Morphogenic slippage and coherence transitions

Morphogenic slippage depends directly on:

- attractive weightings w ;
- coherence redistributions of the three-phase field;
- of morphogenic memory μ ;
- local morphogenic distances $d_i(r)$;
- and variational conditions imposed by the Lagrangian V_{10} .

Within the variational framework of VLCC V_{10} , morphogenic slippage also plays a fundamental role in the operational definition of the inertial environment T_{env} .

PART III — MORPHOGENIC INERTIAL ENVIRONMENT

III.1 — Definition and operational status of the inertial environment (T_{env})

Within the framework of VLCC V10, the observable inertial properties of a system are not considered to be independent of its local state of morphogenic coherence.

The inertial environment is defined by:

$$T_{env}(r) = \sum_i w_i(r) F_i(r)$$

Or :

- $w_i(r)$ represents the local weighting of the attractor G_i ;
- $F_i(r)$ represents the normalized morphogenic response associated with the attractor G_i .

In the VLCC V10, T_{env} is neither a classical thermodynamic temperature nor an autonomous fundamental field. It corresponds to an effective operational projection describing the local level of morphogenic coherence resulting from the attractorial redistributions of the three-phase field.

The functions:

$$F_i(r)$$

do not represent independent fundamental physical fields but effective responses associated with the morphogenic attractors of the local variational landscape.

III.2 — Effective inertial responses and morphogenic mass

The model thus introduces an effective inertial mass:

$$m_{eff}(r)$$

associated with the local state of coherence of the system in the morphogenic space.

In its general formulation, the VLCC V10 introduces the operational relationship:

$$m_{eff}(r) = m_0 f(T_{env}(r))$$

Or :

- m_0 represents the reference inertial mass of the system;
- $f(T_{env})$ represents an effective morphogenic function describing the local inertial response associated with the coherence environment.

A standardized minimal form can, however, be introduced in the form of:

$$f(T_{env}) = \exp(T_{env})$$

This exponential relationship constitutes a simple operational calibration allowing to represent certain progressive redistributions of morphogenic coherence in the continuous regimes of the model.

III.3 — Effective energy response and inertial reformulation

The model introduces an effective energy dependent on the local morphogenic environment described by T_{env} .

In its general formulation, the VLCC V10 introduces the following relationship:

$$E_{eff}(r) = m_0 c^2 f(T_{env}(r))$$

Or :

- m_0 represents the reference inertial mass;
- c represents the standard relativistic limiting velocity;
- $f(T_{env})$ represents the effective morphogenic function associated with the local inertial environment.

A minimal calibration can be introduced in the form of:

$$f(T_{env}) = \exp(T_{env})$$

This exponential form allows for a simple representation of certain progressive redistributions of coherence in the continuous attractor regimes of the model.

The model thus retains a reading compatible with standard relativistic structures while introducing a continuous morphogenic dependence of local energy responses.

PART IV — VARIATIONAL STRUCTURE AND LAGRANGIAN V10

IV.1 — Variational condition and open framework equation

Within the framework of VLCC V10, observable structures are interpreted as emergent states produced by the variational dynamics of the three-phase field and by local attractorial redistributions associated with morphogenic sliding mechanisms.

The model introduces a general variational condition intended to represent the local morphogenic balance of observable structures within the continuous attractorial landscape of the VLCC.

The general morphogenic condition of VLCC V10 is defined by:

$$CVLCC(r) = \sum_i w_i(r) \Phi_i(r) \approx 0$$

Or :

- $w_i(r)$ represents the local weighting of the attractor G_i ;
- $\Phi_i(r)$ represents the variational contribution associated with the attractor G_i .

This relationship constitutes the open framework equation of the VLCC V10. It describes the local conditions of variational stabilization associated with the morphogenic coherence redistributions of the system.

In this approach, observable structures correspond to states of relative minimization of weighted attractor variational contributions.

Morphogenic transitions then appear as progressive redistributions of attractorial weights within the continuous variational landscape of the model.

The relationship:

$$CVLCC(r) \approx 0$$

This corresponds to a condition of effective morphogenic stability describing the locally stabilized regimes of the observable system.

The VLCC V10 then introduces a local variational condition associated with fluctuations in the morphogenic landscape:

$$\delta CVLCC(r) \rightarrow 0$$

This relationship describes the local variational convergence of the system towards a state of relative attractorial stabilization.

In this formulation:

- $CVLCC(r) \approx 0$ describes the stabilized morphogenic structure of the system;
- while $\delta CVLCC(r) \rightarrow 0$ describes the local dynamics of variational convergence towards this stabilization.

This distinction allows us to:

- to preserve the open character of the variational landscape;
- to differentiate between stabilized states and convergence dynamics;
- and to maintain a continuous reading of morphogenic transitions.

The open-ended framework equation plays a direct role:

- in sliding mechanisms;
- in attractor transitions;
- in effective inertial responses;
- in the definition of the Lagrangian V10;
- and in the mechanisms of morphogenic revealability.

The VLCC V10 thus introduces a continuous variational structure in which observable regimes emerge from the local minimization of weighted attractor contributions within the three-phase field.

IV.2 — Attractive variational contributions

Within the framework of VLCC V10, variational contributions:

$\Phi_i(r)$

intervene as local components of the morphogenic variational condition associated with the continuous attractors of the model.

These contributions describe the local discrepancies between observable structures and the effective coherent responses produced by morphogenic attractors.

In its minimal operational formulation, VLCC V10 introduces:

$$\Phi_i(r) = O_{obs}(r) - F_i^*(r)$$

Or :

$(O_{obs}(r))$ represents the local observable structure under consideration;

$(F_i^*(r))$ represents the normalized morphogenic response associated with the attractor (G_i) .

This formulation expresses the fact that observable structures emerge from a variational equilibrium between the attractorial responses of the three-phase field and the local configurations of the system.

The functions:

$$F_i^*(r)$$

correspond to effective coherent responses derived from attractor weightings and morphogenic sliding mechanisms.

The VLCC V10 thus considers that physical observables correspond to states of relative minimization of variational deviations:

$$\Phi_i(r)$$

weighted by the local attractor dynamics of the system.

This open variational structure allows us to describe emergent observable states produced by a continuous redistribution of morphogenic coherence.

Variational contributions come into play directly:

in the construction of the open-ended framework equation; in the mechanisms of attractor stabilization; in the effective inertial responses; in morphogenic transitions; and in the variational structure of the Lagrangian V10.

The observable structures thus appear as local configurations of coherence resulting from a progressive minimization of attractorial variational gaps within the morphogenic space.

IV.3 — Variational Lagrangian V10

Within the framework of VLCC V10, the morphogenic dynamics of the observable system is described from an open variational formalism intended to represent the local redistributions of coherence of the three-phase field.

The model introduces an effective variational Lagrangian associated with the mechanisms of attractor stabilization and local morphogenic responses of the system.

In some intermediate formulations of the VLCC, the Lagrangian had been simplified into a reduced form:

$$LV10 = LGR + LT + LM - \lambda V^2$$

This writing allowed for the introduction of an initial variational stabilization structure but did not explicitly describe the dynamics of the weighted attractorial contributions of the model.

VLCC V10 now introduces a complete variational formulation of the Lagrangian:

$$LV10 = LGR + LT + LM - \Lambda V [\sum_i w_i(r) \Phi_i(r)]^2$$

Or :

- LGR represents the effective relativistic geometric contribution;
- LT represents the contribution associated with the three-phase field;
- LM represents the morphogenic contribution linked to attractors and sliding mechanisms;
- ΛV represents the variational coefficient of morphogenic stabilization;
- $w_i(r)$ represents the local weighting of the attractor G_i ;
- $\Phi_i(r)$ represents the variational contribution associated with the attractor G_i .

This formulation constitutes the main variational core of the VLCC V10. It explicitly introduces a continuous minimization structure based on local attractorial redistributions of the morphogenic landscape.

The term:

$$[\sum_i w_i(r) \Phi_i(r)]^2$$

plays a central role in stabilizing observable structures. It describes the mechanisms of variational minimization associated with local equilibria of morphogenic coherence.

In this approach, physical structures emerge as states of relative stabilization produced by the continuous dynamics of weighted attractor contributions.

The coefficient of variation:

$$\Lambda V$$

controls the intensity of morphogenic stabilization mechanisms and intervenes directly in:

- attractor transitions;
- the sliding mechanisms;
- the redistributions of coherence;
- the effective inertial responses;
- and the overall variational dynamics of the system.

This reformulation allows us to:

- to preserve the morphogenic continuity of the model;
- to avoid rigid transitions between regimes;
- to introduce hybrid states;
- and to stabilize the continuous attractors of the variational landscape.

The VLCC V10 thus retains an open variational logic in which observable structures are no longer interpreted as absolute geometric solutions but as emergent states produced by local mechanisms of attractorial stabilization.

The V10 variational Lagrangian thus constitutes one of the major formal cores of the model and represents the main mathematical structure for stabilizing the morphogenic landscape of the VLCC.

IV.4 — Explicit morphogenic memory

In previous formulations of VLCC, the morphogenic contribution:

L_M

intervened mainly as an effective stabilization term associated with the attractorial dynamics of the three-phase field.

VLCC V10 now introduces an explicit formulation of the memory term within the variational Lagrangian in order to formalize the coherent persistence of local attractorial states.

In this approach, morphogenic memory corresponds to a continuous dynamic of relative stabilization of attractor weights.

The term "memory of the Lagrangian" is introduced in the form:

$$L_M = -\eta \sum_i [w_i(r) - \mu w_i(r-\Delta r) - (1-\mu)w_i^0(r)]^2$$

Or :

$w_i(r)$ represents the current local weighting of the attractor (G_i); $w_i(r)$ represents the morphogenic weighting inherited from the previous state;
 $w_i^0(r)$ represents the reference local attractor configuration;
 η represents the morphogenic memory coefficient;
 μ represents the variational coefficient of memory stabilization.

This formulation expresses the fact that observable structures tend to maintain local coherent continuity despite the attractorial redistributions produced by morphogenic sliding mechanisms.

The term:

$$w_i(r-\Delta r)$$

introduces a dynamic persistence of previously stabilized regimes.

The term:

$$(1-\mu)w^0_i(r)$$

introduces a local attractorial recall contribution associated with the reference morphogenic configuration of the system.

This structure constitutes an effective variational dynamic describing the progressive stability of local coherent regimes.

This reformulation allows, in particular:

to stabilize continuous attractor transitions; to
avoid certain variational discontinuities; to
formalize soft memory mechanisms;
and to explicitly integrate morphogenic persistence into the structure of the Lagrangian V10.

The VLCC V10 thus considers that observable structures emerge from a continuous variational compromise between attractorial redistribution, coherent memory and morphogenic sliding dynamics.

The term explicit memory therefore constitutes one of the main internal mechanisms for stabilizing the morphogenic variational landscape of the model.

IV.5 — VLCC V10 Operational Architecture

The operational architecture of the VLCC V10 is based on a continuous variational chain linking:

the three-phase field;
continuous morphogenic attractors; the
morphogenic variator;
the inertial environment T_{env} ;
local variational contributions; the open
framework equation;
and the variational Lagrangian V10.

In this structure, observable responses emerge from local redistributions of morphogenic coherence produced by the attractorial dynamics of the system.

The VLCC V10 thus describes physical structures as states of local variational stabilization resulting from the dynamic equilibrium between:

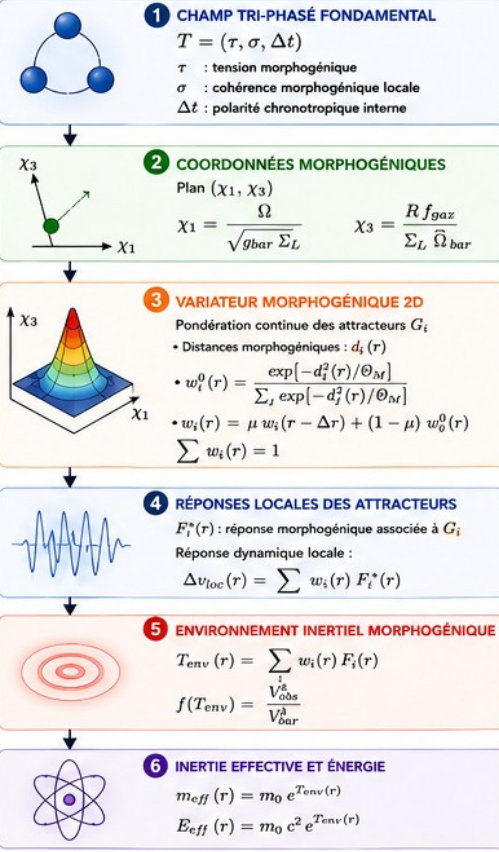
attractor weightings; sliding
mechanisms; effective inertial
responses;
and the mechanisms of morphogenic memory.

This architecture constitutes the general operational core of the VLCC V10 and serves as a support for the cosmological and variational developments of the model.

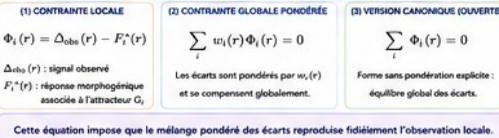
VLCC V10 — ARCHITECTURE OPÉRATEUR MORPHOGÉNIQUE ET DYNAMIQUE RELATIONNELLE

Le présent observable émerge d'une superposition pondérée de régimes morphogéniques attracteurs, modulée par la mémoire du passé (t_1) et orientée par le futur (t_1') dans le présent actif (t_2).

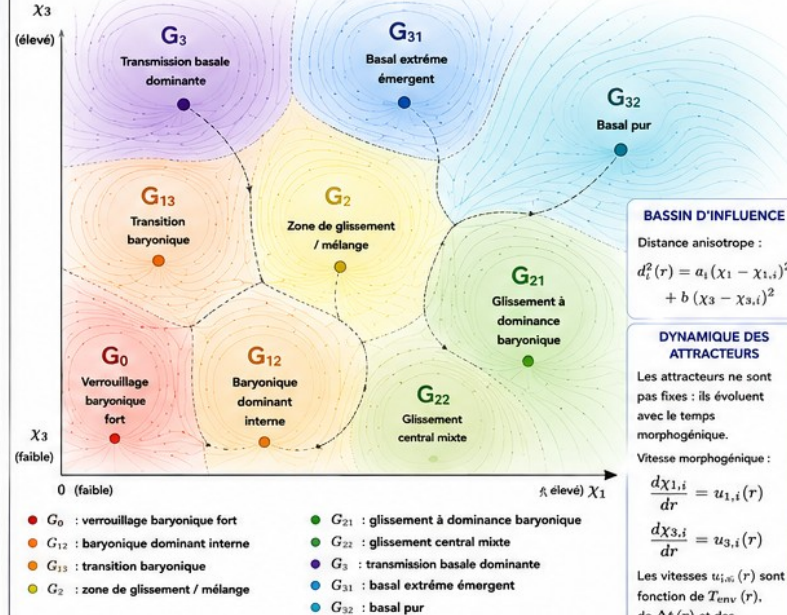
A. CHAÎNE OPÉRATEUR GLOBALE — DYNAMIQUE RELATIONNELLE



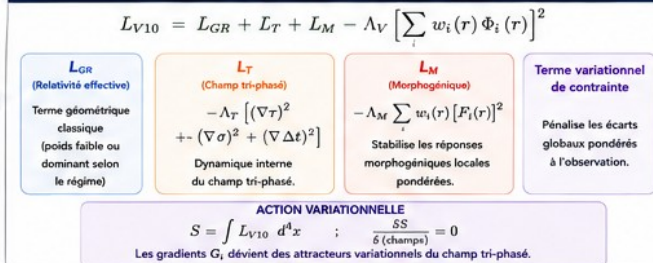
D. ÉQUATION-CADERE OUVERTE — CONTRAINTES GLOBALES



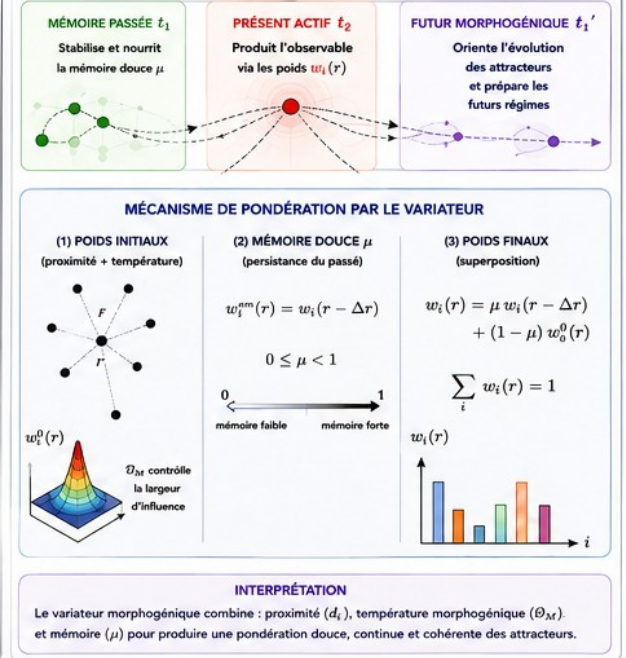
B. PLAN MORPHOGÉNIQUE (χ_1, χ_3) — ATTRACTEURS ET BASSINS D'INFLUENCE DYNAMIQUES



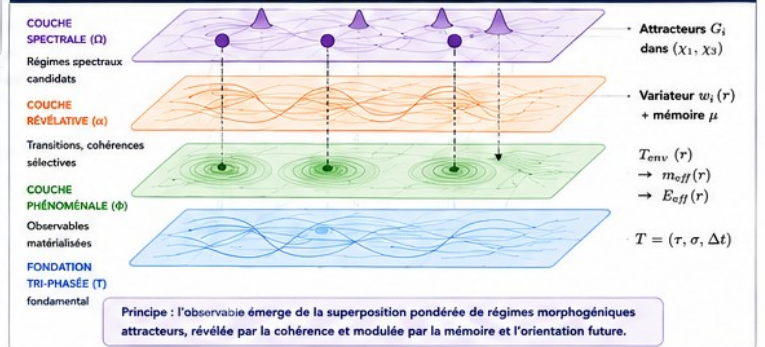
E. LAGRANGIEN VLCC V10 — STRUCTURE VARIATIONNELLE



C. TRIPLÉT MORPHOGÉNIQUE ET VARIATEUR — LECTURE INTÉGRÉE



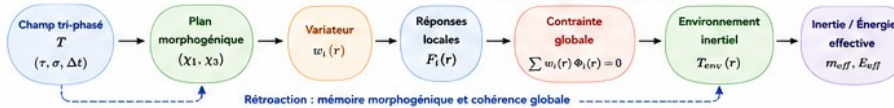
F. ARCHITECTURE STRATIFIÉE DU VLCC V10



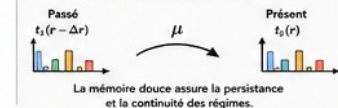
LÉGENDE RAPIDE

w_i : poids morphogénique de l'attracteur G_i
 F_i^* : réponse morphogénique associée à G_i
 r : position locale
 T_M : échelle morphogénique ($-\theta_M$)
 μ : mémoire douce ($0 \leq \mu < 1$)
 T_{env} : environnement inertiel morphogénique
 m_{eff} : masse effective
 Δt : tension chronotropique (polarité)

RÉSUMÉ DYNAMIQUE GLOBAL



MÉMOIRE MORPHOGÉNIQUE



CONSÉQUENCE OPÉRATEUR

Le VLCC V10 est un système variationnel morphogénique où les attracteurs de gradients, en mouvement dans le plan (χ_1, χ_3) , sont sélectionnés en continu par un variateur mémoriel. L'observable (θ_{obs}) émerge d'un mélange pondéré cohérent entre mémoire du passé (t_1) et orientation du futur (t_1').

PART V — REVEABILITY AND MORPHOGENETIC LIMITS

V.1 — Morphogenic revealability and observable coherence

Within the framework of VLCC V10, observable physical structures are not considered to be solely defined by their geometric existence in classical spacetime.

The model introduces an operational distinction between geometric existence and morphogenic revealability.

This distinction is one of the direct consequences of the variational dynamics of the three-phase field. A system can thus retain a partial geometric structure while exhibiting a low level of local morphogenic coherence, limiting its revealability in the observable regime.

The VLCC V10 interprets physical observables as emergent states dependent on the local level of attractor coherence of the system.

Morphogenic revealability depends directly on:

of local coherence σ ;
attractor weightings w_i ; sliding
mechanisms; variational
transitions;
and the morphogenic inertial environment Te_{nv} .

In this formulation, observable structures correspond to coherence regimes sufficiently stabilized to produce local operational revealability. VLCC V10 thus considers that the different morphogenic regimes possess distinct levels of revealability:

Highly coherent regimes correspond to highly stabilized structures exhibiting high revealability;

Transitional regimes correspond to hybrid structures in which attractorial redistributions produce local variations in revealability;

Weak coherent regimes correspond to residual coherence structures for which revealability becomes partial, diffuse, or asymptotic.

This approach allows for a continuous reading of physical observables without imposing a rigid separation between the existence and non-existence of morphogenic structures.

In VLCC V10, revealability is therefore not interpreted as an absolute property but as an emergent quantity dependent on the local coherence conditions of the system.

This reading has a direct impact:

in attractor transitions; in slippage mechanisms; in coherence redistributions; in effective inertial responses;
and in the variational structures of the model.

The VLCC V10 thus introduces a continuous logic of morphogenic revealability in which physical structures gradually emerge according to their local level of attractive stability.

The asymptotic zones of the model can then be interpreted as regimes of residual coherence in which morphogenic structures persist partially without producing complete observable revealability.

Within the operational framework of the VLCC V10, morphogenic revealability thus constitutes a fundamental dynamic property of the observable system and participates directly in the continuous organization of the attractor regimes of the model.

V.2 — Residual coherence and asymptotic structures

Within the framework of VLCC V10, weak coherent regimes are not interpreted as domains of total absence of morphogenic structure but as states of residual coherence associated with low operational revealability.

The attractors associated with the domains:

G3;

G31;

and G32

correspond to structures of weak coherence in which morphogenic responses remain partially organized without producing complete observable stability.

The VLCC V10 thus considers that physical structures can persist in the form of residual coherence even when their operational revealability becomes greatly reduced.

In this approach, boundary regimes correspond to domains in which:

the attractor weightings become diffuse;
sliding mechanisms dominate local stabilizations; variational
redistributions become highly dispersed;
and morphogenic revealability tends towards residual or asymptotic forms.
Asymptotic structures appear when:
sliding mechanisms become dominant; attractor
stabilizations become insufficient;
variational redistributions become highly dispersed; and local
coherence tends towards regimes of low revealability.

In asymptotic regimes:

morphogenic attractors remain partially active; attractor
weightings become unstable;
the effective inertial responses become diffuse;
and the mechanisms of revealability gradually lose their local stability. This
reading allows us to introduce a continuity between:
strongly coherent structures;
transitional regimes;
residual coherences; and
asymptotic domains.

The VLCC V10 thus avoids the introduction of absolute boundaries between observable and unobservable. Physical structures become progressive states of coherence and revealability dependent on the local variational conditions of the system.

Within the variational framework of the model, boundary regimes appear as zones of morphogenic dispersion characterized by:

a low local coherence σ ;
unstable attractor redistributions;
prolonged shifts;
and incomplete variational stabilization.

This approach also allows us to interpret certain weak cosmological structures as residual manifestations of morphogenic coherence rather than as complete absences of physical structure.

The VLCC V10 thus introduces a continuous reading of morphogenic limits in which asymptotic states correspond to dispersed forms of variational stabilization.

This reformulation has a direct impact:

in attractor transitions; in slippage
mechanisms; in coherence
redistributions; in variational
responses;
and in the overall dynamics of the three-phase field.

The model thus maintains a continuous structure compatible with:

gradual transitions; partial
stabilizations; residual
revealabilities;
and the asymptotic dynamics of the observable system.

The regimes of residual coherence thus constitute one of the natural consequences of the open variational dynamics of the VLCC V10 and participate directly in the continuous organization of the morphogenic landscape of the model.

PART VI — OPERATIONAL EXTENSIONS AND MODEL STATUS

VI.1 — Morphogenic continuity and cosmological structures

Within the framework of VLCC V10, observable cosmological structures are not interpreted as isolated objects evolving in a passive spacetime but as local regimes of morphogenic coherence resulting from the variational dynamics of the three-phase field.

The model thus introduces a continuous reading of cosmological structures in which:

observable objects; attractor
transitions; coherence
redistributions; and asymptotic
regimes
belong to the same overall morphogenic dynamic.

The VLCC V10 thus considers that cosmological structures emerge progressively according to their local level of variational stabilization.

In this formulation:

Strongly coherent regimes correspond to strongly stabilized structures; transitional regimes correspond to areas of morphogenic redistribution; and weakly coherent regimes correspond to areas of residual coherence and diffuse revealability.

The observable structures depend on the following:

attractor weightings; slippage
mechanisms; coherence
transitions;
of the morphogenic inertial environment;
and the local variational structure of the three-phase field.

In this approach, cosmological transitions correspond to progressive migrations in morphogenic space rather than absolute breaks in the system.

VLCC V10 thus maintains a continuous logic compatible with:

local redistributions of coherence;
hybrid regimes;
asymptotic structures;
and progressive attractor stabilizations.

This reformulation also allows us to interpret certain diffuse cosmological structures as partial manifestations of morphogenic coherence rather than as anomalies requiring fundamental breaks in the physical framework.

The model does not seek to replace standard cosmological descriptions but to offer a complementary operational reading based on the variational dynamics of morphogenic regimes.

Within the framework of VLCC V10:

observable structures become dependent on their local state of coherence; cosmological transitions become continuous variational phenomena; and physical responses appear as emergent manifestations of the attractorial dynamics of the three-phase field.

Cosmological structures thus appear as emergent dynamic configurations organized within a continuous variational landscape of morphogenic coherence.

VI.2 — Operative status and limitations of the VLCC V10

The VLCC V10 constitutes a speculative operational structure intended to propose a continuous variational reformulation of the morphogenic regimes associated with the present observable.

The model should not be interpreted as a finalized physical theory nor as a direct replacement for the fundamental frameworks currently used in cosmology and theoretical physics.

In its current form, VLCC V10 primarily offers:

an open variational structure; a
continuous attractor dynamic;
a morphogenic reading of the observable present;
and an operational reformulation of local inertial responses.

This approach primarily aims to provide a coherent conceptual framework that allows for:

to unify certain internal structures of the model; to
stabilize the operational formulations;
and to clarify the morphogenic coherence mechanisms introduced in previous versions of the VLCC.

The model thus retains an essentially exploratory status.

The VLCC V10 does not claim to:

replace standard general relativity;
propose a complete quantum theory;
nor provide a definitive description of the fundamental cosmological structures.

The relationships introduced in this work should be understood as effective operational structures intended to represent certain variational dynamics associated with the morphogenic regimes of the model.

Especially :

morphogenic attractors; the
inertial environment T_{env} ; effective
inertial responses; the open-ended
framework equation;
and the Lagrangian V10
correspond to components of an operational framework that is undergoing conceptual
stabilization.

The open nature of VLCC V10 is intentionally one of the central principles of the model.
This openness allows:

to avoid a premature rigidification of mathematical structures; to
preserve variational continuity;
and to maintain the compatibility of the framework with possible future reformulations.

The model thus retains an evolutionary logic in which certain components may be:

revised;
recalibrated;
simplified;
or reformulated according to future developments in the theoretical framework.

In this approach, the mathematical structures proposed in this work should be considered as operational formalization tools and not as definitively validated closed physical descriptions.

The VLCC V10 thus represents an attempt at a coherent structuring of a variational morphogenic framework based on:

the three-phase field;
continuous attractors; sliding
mechanisms;

the redistributions of coherence;
and the revealability dynamics of the observable present.

This reformulation primarily aims to stabilize the operational core of the model in order to allow:

more rigorous analyses;
future mathematical extensions;
and possible conceptual confrontations with certain complex cosmological phenomena.

The status of VLCC V10 thus remains that of an operational speculative framework in the process of formalization, the internal coherence of which currently constitutes the main objective of theoretical development.

GENERAL CONCLUSION

The present work has proposed a variational reformulation of the VLCC based on a continuous dynamics of the morphogenic structures of the present observable.

This evolution of the model, designated as VLCC V10, was primarily aimed at stabilizing the operational core of the theoretical framework while clarifying several structures introduced in previous formulations.

The VLCC V10 now relies on a continuous organization of morphogenic regimes within an attractor space defined by local redistributions of the three-phase field. The former morphogenic gradients are thus reformulated as continuous attractors associated with different levels of morphogenic coherence and linked by variational sliding mechanisms.

This reformulation leads to the introduction of:

- a morphogenic variator designed to weight continuous attractors; an
- effective inertial environment T_{env} describing local coherence responses;
- an open variational condition;
- as well as a V10 Lagrangian explicitly integrating the mechanisms of attractor stabilization and morphogenic memory.

The VLCC V10 thus interprets observable structures as emergent states of local coherence produced by the variational dynamics of the three-phase field rather than as strictly geometric configurations independent of their morphogenic context.

In its current formulation, the model retains a speculative operational status. The structures introduced in this work are primarily formalization tools designed to organize the internal coherence of the theoretical framework and to stabilize its continuous variational architecture.

The objective of this work was therefore not to propose a closed cosmological theory, but to construct a more coherent, more compact and more stable operational structure than previous versions of the VLCC.

This reformulation now opens up several development perspectives regarding:

- the mathematical study of the variational attractorial landscape;
- the dynamics of asymptotic regimes of residual coherence; the
- mechanisms of morphogenic revealability;
- as well as the variational properties associated with effective inertial responses.

The VLCC V10 thus constitutes an attempt at progressive stabilization of a variational morphogenic framework in which the present observable is interpreted as an emerging dynamic structure of local coherence.

Appendix A — Glossary

A.1 — Fundamental Structures

Coherent three-phase field

Fundamental coherent structure of the VLCC defined by:

$$T = (\tau, \sigma, \Delta t)$$

The three-phase field describes the coupled dynamics of morphogenic memory, local coherence, and morphogenic slippage associated with the present observable.

Revealability

Local capacity of a morphogenic state to emerge in the observable present in a coherent and stabilized form.

Present observable

Local regime of stabilized coherence in which morphogenic structures become physically revealable.

Morphogenic field

Continuous coherent structure derived from the three-phase field and associated with the dynamics of slowing down, memory and morphogenic organization of the system.

Morphogenic coherence

Local level of coherent stabilization of the three-phase field.

Morphogenic coherence controls the stability of revealable structures within the present observable.

A.2 — Coherent Dynamics

Morphogenic slippage

Coherent voltage variational dynamics associated with revealability deviations of the three-phase field.

The slip represents the continuous dynamics of local morphogenic transitions.

Sweet memory

Local coherent persistence associated with previous attractor states of the system.

Soft memory is formalized in the V10 framework through the variational coefficient μ .

Morphogenic tension

Local dynamics of coherent stress associated with slip variations and attractor transitions.

Chronotropic polarity

Internal coherent asymmetry of the morphogenic triplet defined by:

$$\Delta t = t1' - t1$$

Chronotropic polarity describes the local orientation of the morphogenic dynamics of the system.

Chronotropic dynamics

Continuous coherent migration between evolutionary tension, coherent present, and morphogenic memory:

$$t1' \rightarrow t2 \rightarrow t1$$

This dynamic constitutes the operational support of the chronotropic arrow of the VLCC V10.

A.3 — Attractive structures**Morphogenic attractant**

A dominant coherent region towards which certain morphogenic slowdown dynamics locally converge.

VLCC V10 attractors are defined as continuous structures and not as compartmentalized discrete states.

Morphogenic gradient

Continuous coherent gradation used to describe the dominant regimes of morphogenic slowdown within the tri-phase field.

Within the V10 framework, the gradient does not designate a separate physical object, but a coherent reading scale associated with morphogenic attractors.

Morphogenic slowdown region

Local coherent zone characterized by a dominant regime of attractive slowdown.

This concept represents the continuous operational reformulation of the old gradients.

compartmentalized morphogenetics.

Attractiveness superposition

Local coherent regime in which several morphogenic attractors contribute simultaneously to the system dynamics through continuous variational weightings.

Morphogenic geodesics

Local variational trajectories followed by a system in the morphogenic space (χ^1, χ^3) .

In the VLCC V10, morphogenic geodesics describe attractor stabilization paths associated with local coherence redistributions.

A.4 — Operational Structures

Tenv

Emergent variational projection of morphogenic coherence derived from the continuous dynamics of attractors of the three-phase field.

Tenv is neither an autonomous fundamental field, nor a classical thermodynamic temperature.

Morphogenic Variator

Continuous variational structure introduced in the V10 framework to formalize local attractor weights and morphogenic memory mechanisms.

Morphogenic response

Local coherent response derived from the attractorial dynamics of the three-phase field.

Morphogenic responses are involved in the operational construction of Tenv.

Open coherent dynamics

Continuous variational architecture in which observable responses depend simultaneously on:

- attractive structures;
- memory mechanisms;
- coherent superpositions;
- and local variational dynamics of the three-phase field.

Open framework equation

General variational relationship of the VLCC V10:

$$CVLCC(r) = \sum_i w_i(r) \Phi_i(r) \approx 0$$

This structure describes the local conditions of morphogenic stabilization of the observable system.

Variational Lagrangian V10

General variational structure of the VLCC V10 incorporating:

- attractor dynamics;
- memory mechanisms;
- morphogenic responses;
- and the variational stabilization structures of the three-phase field.

Operating structure

Mathematical and conceptual organization of the VLCC intended to formalize the variational dynamics of the model independently of any definitive ontological interpretation.

A.5 — Chronotropic structures**Morphogenic triplet**

Fundamental chronotropic structure of the VLCC defined by:

$$(t_1, t_2, t_1')$$

The morphogenic triplet describes the dynamic relationships between morphogenic memory, coherent present, and evolutionary tension.

Morphogenic invariant

Local condition for chronotropic stability defined by:

$$I = t_2^2 - t_1 t_1'$$

This structure describes the admissible dynamic coherence conditions of the present observable.

Present consistent

Dynamic structure associated with t_2 .

The coherent present does not correspond to a specific moment but to a local regime of morphogenic stability.

Chronotropic arrow

Emergent coherent orientation produced by the internal dynamics of the morphogenic triplet.

In VLCC V10, the arrow of time results from a continuous morphogenic migration and not from a geometric orientation imposed a priori.

Residual coherence

Asymptotic morphogenic regime in which certain coherent structures persist locally despite low observable revealability.

A.6 — Photonicotemporal structures**Closed photonic spectrum**

Coherent structure of the photonicotemporal field in which revealing, non-revealing and retro-revealing states are organized according to a continuous morphogenic dynamic.

Zero-frequency photon

Basal photonicotemporal mode associated with a frequency:

$$\nu = 0$$

In the VLCC V10, this mode does not correspond to a classical observable photon but to a coherent state not revealing the closed spectrum.

 $\mathcal{U}$ — Non-revealing photonicotemporal state

State associated with the t_2 component of the morphogenic triplet.

$\mathcal{U}$ designates the operating regime of the non-revealing basal mode at zero frequency. It is neither an autonomous physical particle nor an additional field, but a coherent structure associated with the non-revealing present.

non-revealing basal mode

Photonicotemporal regime of minimal coherence associated with the pre-revealability structures of the three-phase field.

Photonicotemporal structure

Coherent architecture linking:

$$(\alpha, \mathcal{U}, \alpha') \leftrightarrow (t_1, t_2, t_1')$$

This structure formalizes the operational correspondences between photonicotemporal dynamics and chronotropic structures of the VLCC V10.

Appendix B — Detailed mathematical structures of the VLCC V10

B.1 — Fundamental three-phase field

$$T = (\tau, \sigma, \Delta t)$$

B.2 — Chronotropic polarity

$$\Delta t = t1' - t1$$

B.3 — Morphogenic triplet

$$(t1, t2, t1')$$

B.4 — Morphogenic invariant

$$I = t2^2 - t1 \ t1'$$

$$I > 0$$

B.5 — Chronotropic Dynamics

$$t1' \rightarrow t2 \rightarrow t1$$

B.6 — Condition for attractor normalization

$$\sum_i w_i(r) = 1$$

B.7 — Morphogenic Variator

$$w_i(r) = \mu \ w_i(r - \Delta r) + (1 - \mu) \ [\exp(-d_i^2(r) / \Theta M) / \sum_j \exp(-d_j^2(r) / \Theta M)]$$

B.8 — Morphogenic distance

$$d_i^2(r) = a_i(\chi^1 - \chi_{1,i})^2 + b_i(\chi^3 - \chi_{3,i})^2$$

B.9 — Morphogenic Plan

$$(\chi^1, \chi^3)$$

$$\chi^1 = \Omega / \sqrt{g_{\text{bar}} \Sigma L}$$

$$\chi^3 = (R \ f_{\text{gaz}}) / (\Sigma L \ \Omega_{\text{bar}})$$

B.10 — Morphogenic inertial environment

$$T_{\text{env}}(r) = \sum_i w_i(r) \ F_i^*(r)$$

B.11 — Effective Inertial Response

$$m_{eff}(r) = m_0 f(Tenv(r))$$

$$f(Tenv) = \exp(Tenv)$$

B.12 — Effective energy response

$$E_{eff}(r) = m_0 c^2 f(Tenv(r))$$

B.13 — Open framework equation

$$CVLCC(r) = \sum_i w_i(r) \Phi_i(r) \approx 0$$

$$\delta CVLCC(r) \rightarrow 0$$

B.14 — Local variational contribution

$$\Phi_i(r) = O_{obs}(r) - F_i^*(r)$$

B.15 — Local variational structure

$$[\sum_i w_i(r) \Phi_i(r)]^2$$

B.16 — Variational Lagrangian V10

$$LV10 = LGR + LT + LM - \lambda V [\sum_i w_i(r) \Phi_i(r)]^2$$

B.17 — Photonicotemporal structure

$$(\alpha, \mathcal{U}, \alpha')$$

$$(\alpha, \mathcal{U}, \alpha') \leftrightarrow (t_1, t_2, t_1')$$

B.18 — Non-revealing basal mode

$$v(\mathcal{U}) = 0$$

$$m(\mathcal{U}) = 0$$

B.19 — Overall chronotropic structure

Present observable = weighted superposition of attractor morphogenic regimes

Appendix C — Table of morphogenic attractors

Attractor	Phase	Diet morphogenic	Operational nature dominant	Level of consistency	Main morphogenic function
G1	Phase I	Strong diet stabilized	Stabilization coherent maximum	Very strong	highly revealable and highly stabilized attractive structure
G1.2	Phase I	Coherent regime stable	Consistency stabilized dynamics	Forte	Maintaining coherent structures local and revealable regimes persistent
G1.3	Phase I	Diet of active structuring	Redistribution coherent active	Strong at intermediate	Morphogenic structuring and controlled attractor transitions
G2	Phase II	Transitional regime consistent	Transition variational dominant	Intermediate	Redistribution of weightings attractiveness and slippage morphogenic
G2.1	Phase II	Sliding regime dominant	Migration attractive continues	Intermediate unstable	Main dynamics of transition continuous morphogenetic
G2.2	Phase II	Attractiveness regime diffuse	Overlay attractive extent	Low to intermediate	Formation of hybrid regimes and redistribution of coherence
G3	Phase III	Coherent regime weak	Residual coherence dominant	Weak	Weakly morphogenic structures revealable
G3.1	Phase III	Consistency regime residual	Stabilization asymptotic partial	Very low	Maintaining coherent structures diffuse and asymptotic
G3.2	Phase III	Limit regime of revealability	Limit morphogenic asymptotic	Residual	Coherent pre-dispersion domain and minimal revealability

General operational reading of attractors

Within the framework of VLCC V10, morphogenic attractors should not be interpreted as strictly separate discrete physical domains, but as continuous regions of variational stability organized in morphogenic space:

$$(\chi^1, \chi^3)$$

Each attractor represents a dominant regime of morphogenic coherence associated with certain local distributions:

- of attractive weightings;
- of morphogenic slippage;
- of chronotropic stability;
- and observable revealability.

Transitions between attractors correspond to progressive redistributions of coherence and not to absolute breaks between distinct physical states.

In the VLCC V10, an observable system generally results from a continuous superposition of weighted attractor contributions and never belongs strictly exclusively to a single morphogenic attractor.

Appendix D — Mathematical Glossary

Symbol	Operational meaning
T	Fundamental three-phase field of the VLCC
τ	Morphogenic tension
σ	Local morphogenic coherence
Δt	Internal chronotropic polarity
t_1	Stabilized morphogenic memory
t_2	Present coherent operational
t_1'	Morphogenic evolutionary tension
I	Morphogenic invariant
χ_1	Morphogenic coordinate associated with the stabilized coherence
χ_3	Morphogenic coordinate associated with slippage and evolving tension
G	Local morphogenic attractant
$w_i(r)$	Local weighting of the G attractor
μ	Morphogenic memory coefficient
$d^2_i(r)$	Quadratic morphogenic distance local
Θ_m	Effective scale of diffusion morphogenic
$F_i^*(r)$	Normalized morphogenic response associated with the G attractor
T_{env}	Morphogenic inertial environment effective
$T_{env}(r)$	Effective local level of coherence morphogenic

m_0	Reference inertial mass
$m_{\text{eff}}(r)$	Morphogenic effective inertial mass
$E_{\text{eff}}(r)$	Morphogenic energy response effective
$f(T_{\text{env}})$	Effective morphogenic function
c	Standard relativistic velocity
$\Phi_i(r)$	variational contribution associated with the G_i attractor
$CVLCC(r)$	General variational condition of the VLCC
LV_{10}	VLCC V10 variational Lagrangian
LGR	effective relativistic contribution
LT	Contribution of the three-phase field
LM	Morphogenic contribution variational
ΛV	Stabilization variational coefficient
α	Photonicotemporal state revealing
α'	Retro-revealing photonicotemporal state
$\bar{U}$	Non-revealing photonicotemporal state
$\nu(\bar{U})$	Photonicotemporal frequency of the mode basal
$m(\bar{U})$	Effective self-mass of the basal mode
Ω	Local coherent rotational magnitude
r	Local position in space morphogenic
$\exp$	exponential calibration function morphogenic
$\approx$	Approximate variational condition of local stability

→

Continuous morphogenic migration

↔

Operational correspondence between
morphogenic structures